

FROM COST OF EQUITY TO COST OF CAPITAL

- The cost of capital is a composite cost to the firm of raising financing to fund its projects.
- In addition to equity, firms can raise capital from debt.
- To get to a cost of capital, you need to
 - Estimate a cost of debt
 - Estimate weights for debt and equity

WHAT IS DEBT?

- **General Rule:** Debt generally has the following characteristics:
 - **Contractual commitment** to make fixed payments in the future
 - The fixed payments are **tax deductible**
 - Failure to make the payments can **lead to either default or loss of control** of the firm to the party to whom payments are due.
- As a consequence, debt should include
 - Any **interest-bearing liability**, whether short term or long term.
 - Any **lease obligation**, whether operating or capital.

ESTIMATING THE COST OF DEBT

- If the firm has bonds outstanding, and the bonds are traded, **the yield to maturity on a long-term, straight (no special features) bond** can be used as the interest rate.
- If the **firm is rated**, use the **rating and a typical default spread** on bonds with that rating to estimate the cost of debt.
- If the firm is not rated,
 - and it has **recently borrowed long term from a bank**, use the interest rate on the borrowing or
 - estimate a **synthetic rating for the company**, and use the synthetic rating to arrive at a default spread and a cost of debt
- The cost of debt has to be estimated in the same currency as the cost of equity and the cash flows in the valuation.

THE EASY ROUTE: OUTSOURCING THE MEASUREMENT OF DEFAULT RISK

- For those firms that have bond ratings from global ratings agencies, I used those ratings:

Company	S&P Rating	Risk-Free Rate	Default Spread	Cost of Debt
Disney	A	2.75% (US \$)	1.00%	3.75%
Deutsche Bank	A	1.75% (Euros)	1.00%	2.75%
Vale	A-	2.75% (US \$)	1.30%	4.05%

- If you want to estimate Vale's cost of debt in \$R terms, we can again use the differential inflation approach we used for the cost of equity:

$$\begin{aligned}
 \text{Cost of debt}_{R\$} &= (1 + \text{Cost of debt}_{US \$}) \frac{(1 + \text{Expected Inflation}_{R\$})}{(1 + \text{Expected Inflation}_{US \$})} - 1 \\
 &= (1.0405) (1.09/1.02) - 1 = 11.19\%
 \end{aligned}$$

A MORE GENERAL ROUTE: ESTIMATING SYNTHETIC RATINGS

- The **rating for a firm can be estimated using the financial characteristics of the firm**. In its simplest form, we can use just the interest coverage ratio:

$$\text{Interest Coverage Ratio} = \text{EBIT} / \text{Interest Expenses}$$

- For the non-financial service companies, we obtain the following:

Company	Operating income	Interest Expense	Interest coverage ratio
Disney	\$10.023	\$444	22.57
Vale	\$15,667	\$1,342	11.67
Tata Motors	Rs 166,605	Rs 36,972	4.51
Baidu	CY 11,193	CY 472	23.72
Bookscape	\$2,536	\$492	5.16

INTEREST COVERAGE RATIOS, RATINGS AND DEFAULT SPREADS- NOVEMBER 2013

<i>Large cap (>\$5 billion)</i>	<i>Small cap or risky (<\$5 billion)</i>	<i>Rating is (S&P/ Moody's)</i>	<i>Spread (11/13)</i>
>8.50	>12.5	Aaa/AAA	0.40%
6.5-8.5	9.5-12.5	Aa2/AA	0.70%
5.5-6.5	7.5-9.5	A1/A+	0.85%
4.25-5.5	6-7.5	A2/A	1.00%
3-4.25	4.5-6	A3/A-	1.30%
2.5-3	4-4.5	Baa2/BBB	2.00%
2.25-2.5	3.5-4	Ba1/BB+	3.00%
2-2.25	3-3.5	Ba2/BB	4.00%
1.75-2.25	2.5-3	B1/B+	5.50%
1.5-1.75	2-2.5	B2/B	6.50%
1.25-1.5	1.5-2	B3/B-	7.25%
0.8-1.25	1.25-1.5	Caa/CCC	8.75%
0.65-0.8	0.8-1.25	Ca2/CC	9.50%
0.2-0.65	0.5-0.8	C2/C	10.50%
<0.2	<0.5	D2/D	12.00%

Disney: Large cap, developed	22.57	→	AAA
Vale: Large cap, emerging	11.67	→	AA
Tata Motors: Large cap, Emerging	4.51	→	A-
Baidu: Small cap, Emerging	23.72	→	AAA
Bookscape: Small cap, private	5.16	→	A-

SYNTHETIC VERSUS ACTUAL RATINGS: RATED FIRMS

- **Disney's synthetic rating is AAA**, whereas its **actual rating is A**. The difference can be attributed to any of the following:
 - Synthetic ratings reflect only the interest coverage ratio whereas actual ratings incorporate all of the other ratios and qualitative factors
 - Synthetic ratings do not allow for sector-wide biases in ratings
 - Synthetic rating was based on 2013 operating income whereas actual rating reflects normalized earnings
- **Vale's synthetic rating is AA**, but the **actual rating for dollar debt is A-**. The biggest factor behind the difference is the presence of country risk, since Vale is probably being rated lower for being a Brazil-based corporation.
- **Deutsche Bank had an A rating**. We will not try to estimate a synthetic rating for the bank. Defining interest expenses on debt for a bank is difficult...

ESTIMATING COST OF DEBT

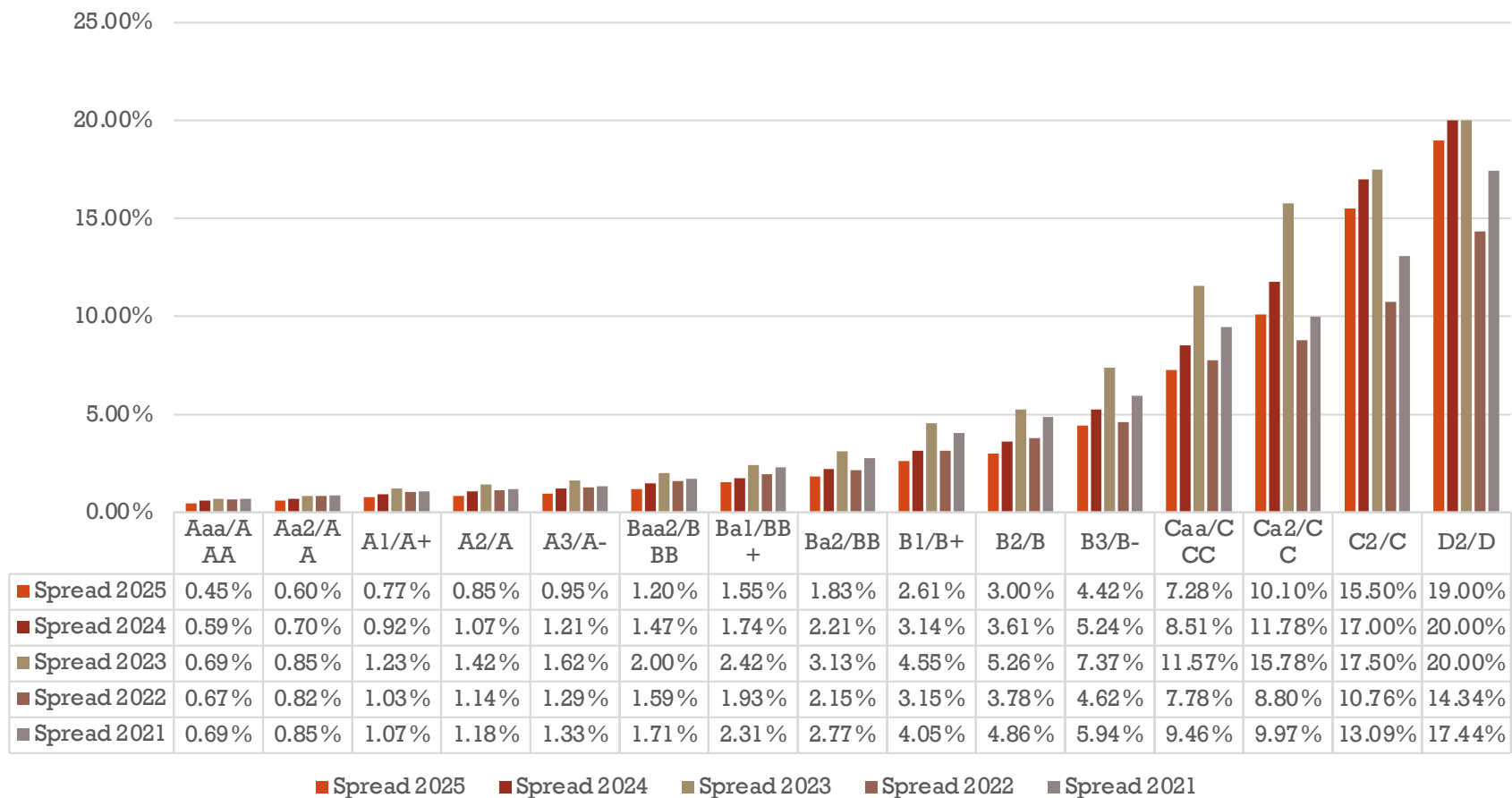
- For Bookscape, we will use the synthetic rating (A-) to estimate the cost of debt:
 - Default Spread based upon A- rating = 1.30%
 - Pre-tax cost of debt = Riskfree Rate + Default Spread = 2.75% + 1.30% = 4.05%
 - After-tax cost of debt = Pre-tax cost of debt (1- tax rate) = 4.05% (1-.40) = 2.43%
- For the three publicly traded firms that are rated in our sample, we will use the actual bond ratings to estimate the costs of debt.

Company	S&P Rating	Risk-Free Rate	Default Spread	Cost of Debt	Tax Rate	After-Tax Cost of Debt
Disney	A	2.75% (US \$)	1.00%	3.75%	36.1%	2.40%
Deutsche Bank	A	1.75% (Euros)	1.00%	2.75%	29.48%	1.94%
Vale	A-	2.75% (US \$)	1.30%	4.05%	34%	2.67%

- For Tata Motors, we have a rating of AA- from CRISIL, an Indian bond-rating firm, that measures only company risk. Using that rating:
 - Cost of debt_{TMT} = Risk free rate_{Rupees} + Default spread_{India} + Default spread_{TMT}
 - = 6.57% + 2.25% + 0.70% = 9.62%
 - After-tax cost of debt = 9.62% (1-.3245) = 6.50%

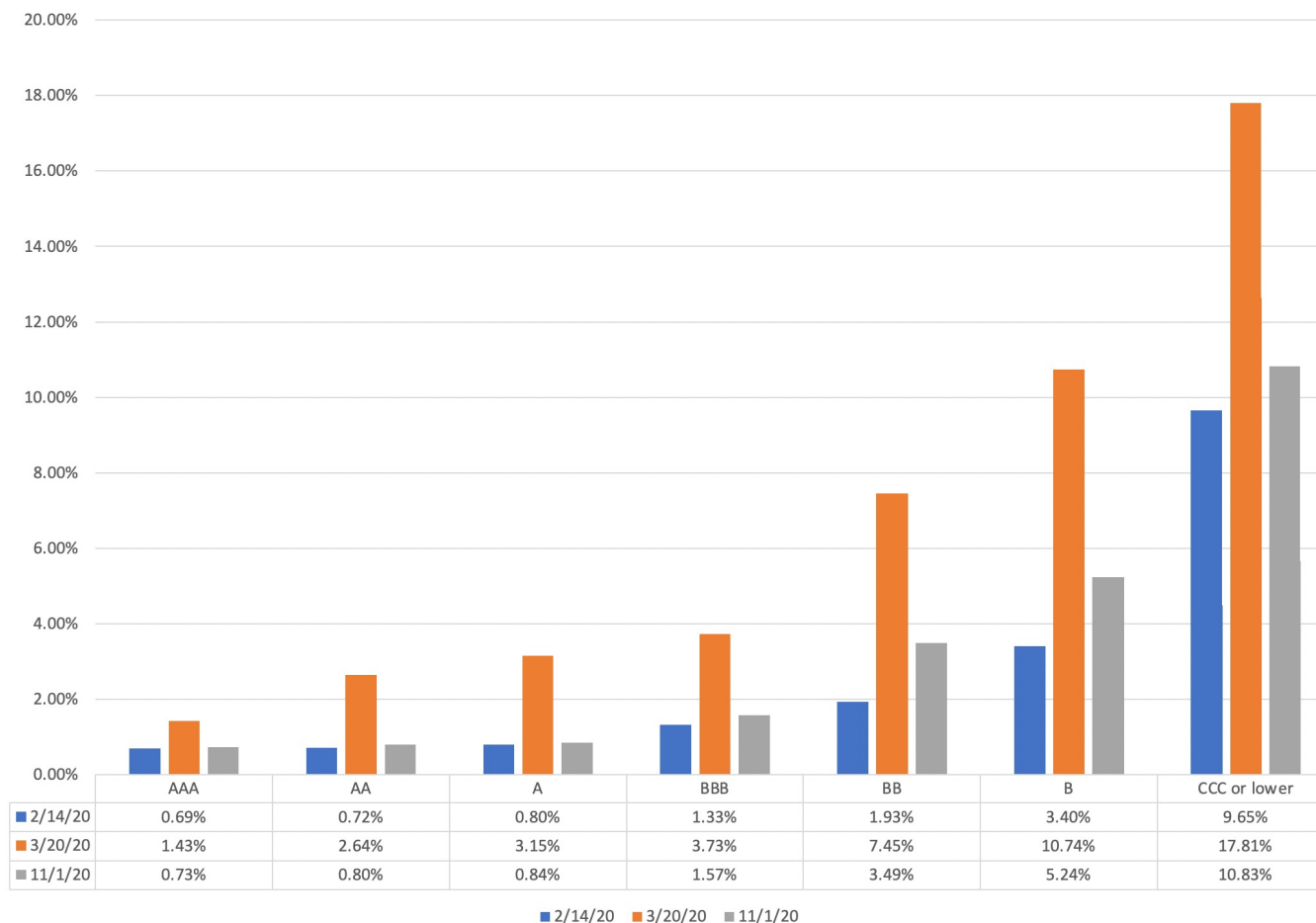
DEFAULT SPREADS – JANUARY 2024

Corporate Bond Default Spreads on January 1, 2025



BUT SOME YEARS ARE VOLATILE: 2020 AS A CASE STUDY..

Corporate Bond Default Spreads: 2/14 - 11/1



APPLICATION TEST: ESTIMATING A COST OF DEBT

- Based upon your firm's current earnings before interest and taxes, its interest expenses, estimate
 - An interest coverage ratio for your firm
 - A synthetic rating for your firm (use the tables from prior pages)
 - A pre-tax cost of debt for your firm
 - An after-tax cost of debt for your firm

COSTS OF HYBRIDS

- **Preferred stock** shares some of the characteristics of debt - the preferred dividend is pre-specified at the time of the issue and is paid out before common dividend -- and some of the characteristics of equity - the payments of preferred dividend are not tax deductible. If preferred stock is viewed as perpetual, the cost of preferred stock can be written as follows:
 - $k_{ps} = \text{Preferred Dividend per share} / \text{Market Price per preferred share}$
- **Convertible debt** is part debt (the bond part) and part equity (the conversion option). It is best to break it up into its component parts and eliminate it from the mix altogether.